ASE2910 Applied Linear Algebra / AUS2910 Fundamental Math for AI
Homework #4

1) *Lower triangular matrices.* Let A be a lower triangular $n \times n$ matrix. Verify the following properties.

   a) If B is a lower triangular $n \times n$ matrix, then the product AB is lower triangular.

   b) The matrix A^k is lower triangular for all positive integers k.

   c) If A is invertible, then A^k is lower triangular for all integers k (positive or negative).

2) *Circular shift matrices.* Let A be the 5×5 matrix:

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

   a) How is Ax related to x? *Hint. See exercise title.*

   b) What is A^5? *Hint. The answer should make sense, given your answer to part (a).*

3) *VMLS Exercises.*

   a) **9.1** *Compartmental system.*

   b) **9.3** *Equilibrium point for linear dynamical system.*

   c) **9.5** *Fibonacci sequence.*

   d) **9.6** *Recursive averaging.*

   e) **10.12** *Norm of matrix product.*

   f) **10.19** *Student group membership.*

   g) **10.22** *Composition of linear dynamical systems.*

   h) **10.25** *Squareroots of the identity.*

   i) **10.32** *Matrix exponential.*

   j) **10.35** *Orthogonal matrices.*